

ECON 1550 International Finance

Price Levels and the Exchange Rate in the Long Run

Law of One Price (LOOP)

- The LOOP holds if for every good i with dollar price P_{US}^i and euro price P_E^i , we have

$$P_{US}^i = E_{\$/\epsilon} \times P_E^i$$

- This is a theory of exchange rate determination:

$$E_{\$/\epsilon} = \frac{P_{US}^i}{P_E^i}$$

Purchasing Power Parity (PPP)

- PPP holds if

$$P_{US} = E_{\$/\epsilon} \times P_E$$

where P_{US} is the US price level and P_E is the euro area price level

- This is also a theory of exchange rate determination:

$$E_{\$/\epsilon} = \frac{P_{US}}{P_E}$$

Relative PPP

- Relative PPP holds if

$$\frac{E_{\$/\epsilon,t} - E_{\$/\epsilon,t-1}}{E_{\$/\epsilon,t-1}} = \pi_{US,t} - \pi_{E,t}$$

where π_t is inflation $\pi_t = P_t/P_{t-1} - 1$

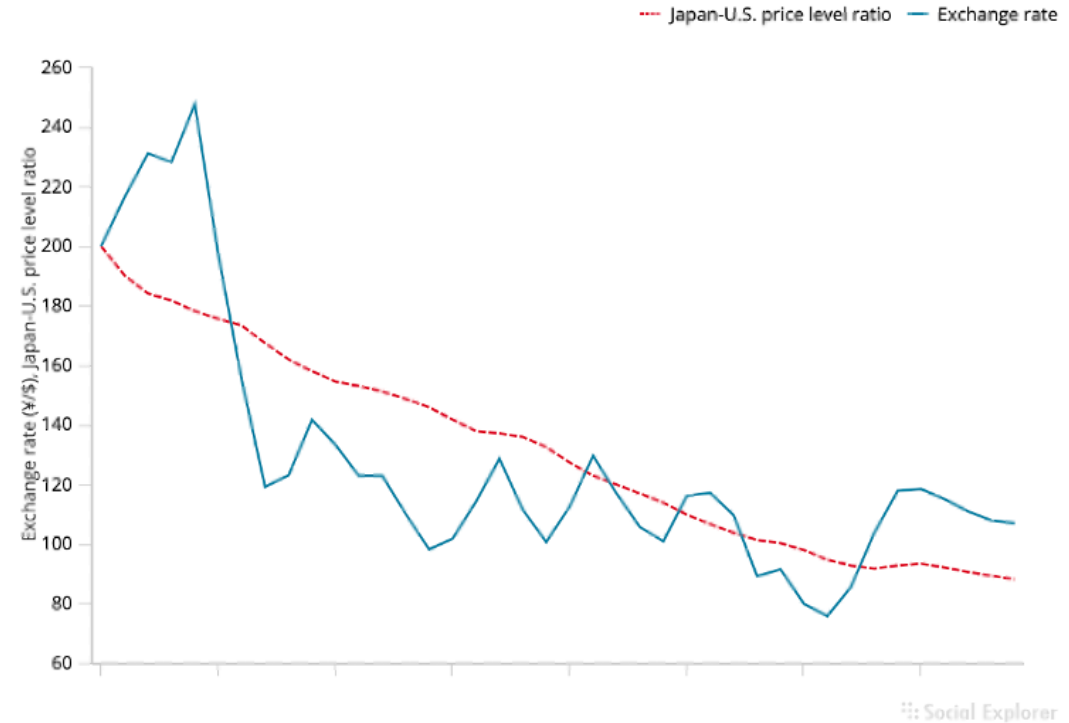
- This is also a theory of exchange rate determination

$$E_{\$/\epsilon,t} = (\pi_{US,t} - \pi_{E,t} + 1)E_{\$/\epsilon,t-1}$$

Relative PPP
does not hold
very well in
the real world

Figure 16-2

The Yen/Dollar Exchange Rate and Relative Japan-U.S. Price Levels, 1980–2019



The graph shows that relative PPP does not track the yen/dollar exchange rate during 1980–2015.

Source: IMF, *International Financial Statistics*. Exchange rates and price levels are end-of-year data.

Problems With PPP

- Price levels of different countries report different baskets of goods
 - E.g. GDP deflator is the basket of goods in GDP for each country
- Deviations from perfectly competitive frictionless markets
 - Transport costs and trade barriers
 - Monopoly power

Expected Relative PPP

- Expected relative PPP (or relative PPP in expectations)

$$\frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}} = \pi_{US}^e - \pi_E^e$$

where π^e is *expected* inflation $\pi^e = P^e/P - 1$

- This is also a theory of exchange rate determination

$$E_{\$/\epsilon} = \frac{E_{\$/\epsilon}^e}{\pi_{US}^e - \pi_E^e + 1}$$

Relationships

- LOOP $\Rightarrow$ PPP $\Rightarrow$ Relative PPP $\Rightarrow$ Expected relative PPP
- Expected relative PPP + UIP $\Rightarrow$ Fisher effect

$$\frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}} = \pi_{US}^e - \pi_E^e \quad \text{and} \quad R_{\$} = R_{\epsilon} + \frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}}$$

imply

$$R_{\$} - R_{\epsilon} = \pi_{US}^e - \pi_E^e$$

Common variables across all models

Exogenous variables

Variable	Description
R^*	Foreign interest rate
M^s, M^{s*}	Money supply
g_M, g_{M^*}	Money supply growth rates

Endogenous variables

Variable	Description	Equation	Type of equation
P	Price level	$M^s/P = L(R, Y)$	Behavioral + eq. condition
P^*	Foreign price level	$M^{s*}/P^* = L(R^*, Y^*)$	Behavioral + eq. condition
π, π^*	Inflation	$\pi = g_M - g_L$	Definition
r^e, r^{e*}	Real interest rates	$r^e = R - \pi^e$	Definition

Model 1: PPP

Exogenous variables

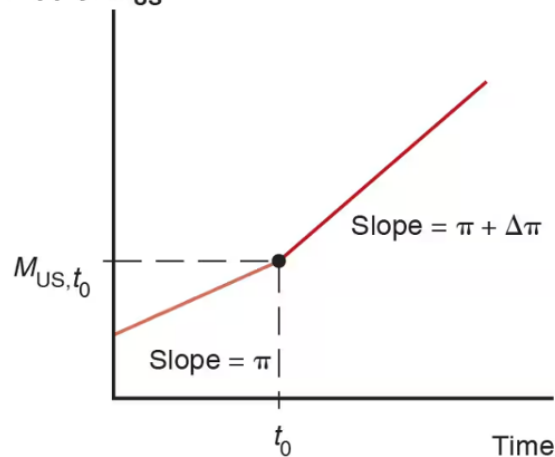
Variable	Description
R	Domestic interest rate
Y, Y^*	Real incomes

Endogenous variables

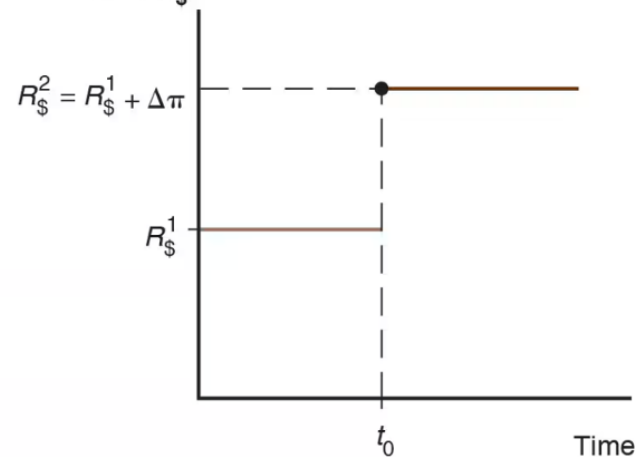
Variable	Description	Equation	Type of equation
E	Exchange rate	$E = P/P^*$	Behavioral

Changes in growth rate of money supply

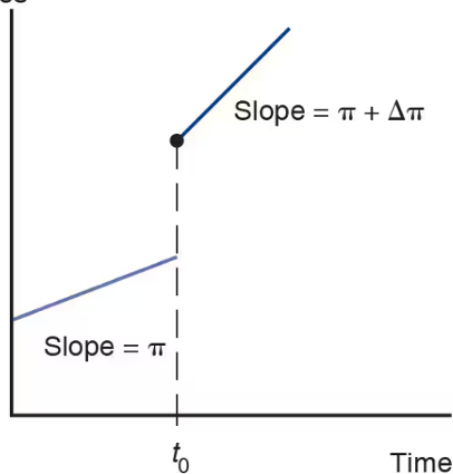
(a) U.S. money supply, M_{US}



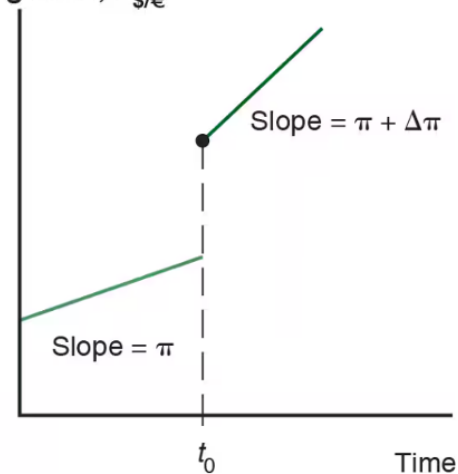
(b) Dollar interest rate, $R_{\$}$



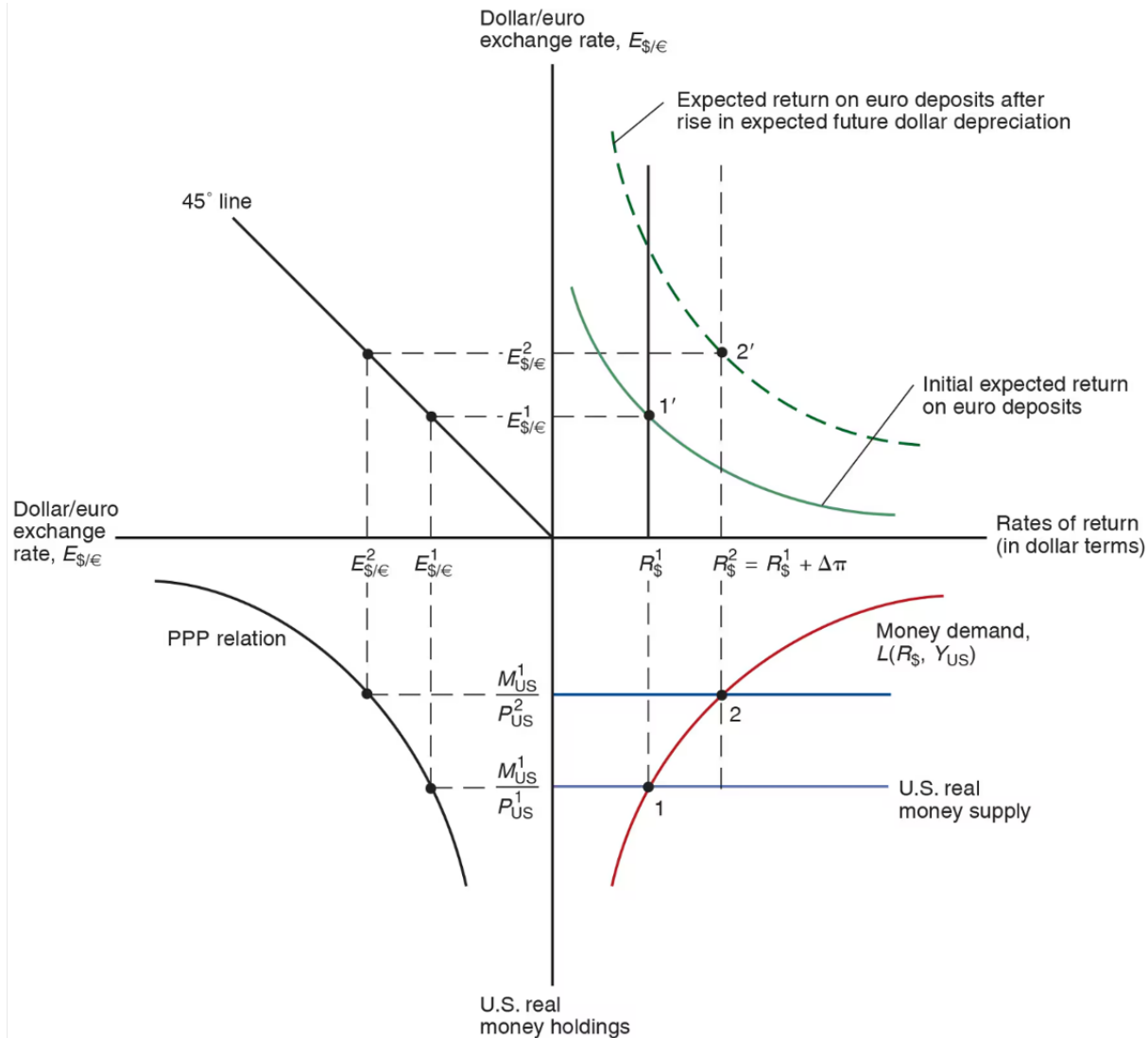
(c) U.S. price level, P_{US}



(d) Dollar/euro exchange rate, $E_{\$/\epsilon}$



Changes in growth rate of money supply



Model 2: PPP^e + UIP

Exogenous variables		Endogenous variables			
Variable	Description	Variable	Description	Equation	Type of equation
Y, Y^*	Real incomes	$E^e/E - 1$	Expected depreciation	$E^e/E - 1 = \pi^e - \pi^{e*}$	Behavioral
		R	Interest rate	$R = R^* + E^e/E - 1$	Eq. condition

Model 3: real E + relative output

- The real exchange rate : $q \equiv E \cdot \frac{P^*}{P}$

$$E = 2 \text{ USD/EUR}$$

$$P = 5 \text{ USD/US-GOODS}$$

$$P^* = 3 \text{ EUR/EURO-GOOD}$$

$$q = 2 \frac{\cancel{\text{USD}}}{\cancel{\text{EUR}}} \times 3 \frac{\cancel{\text{EUR}}}{\text{EURO-GOOD}} \frac{1}{\frac{5 \cancel{\text{USD}}}{\text{US-GOOD}}}$$

$$= \frac{6}{5} \frac{\text{US-GOODS}}{\text{EURO-GOODS}}$$

$E \cdot P^*$ = DOLLAR PRICE OF
FOREIGN GOODS

Model 3: real E + relative output

- Real exchange rate calculation

$$q = E \cdot \frac{P^*}{P} = \frac{6}{5} \frac{\text{US-GOODS}}{\text{EUR-GOODS}}$$

$$(\text{PPP}) : E = \frac{P}{P^*} \quad \Leftrightarrow \quad q = 1$$

$$\Rightarrow E = q \cdot \frac{P}{P^*} = k \cdot P \quad \text{with } k = \frac{q}{P^*}$$

Model 3: real E + relative output

$$E = 1 \text{ USD/EUR}$$

↓

$$E = 2 \text{ USD/EUR}$$

↓

EUR CAN BUY

MORE

US GOODS

Exogenous variables

Variable	Description
RS	Relative output supply
q	Real exchange rate

Endogenous variables

Variable	Description	Equation	Type of equation
Y/Y^*	Relative output	$RD(q) = RS$	Eq. cond.
E	Nominal exchange rate	$E = qP/P^*$	Definition

$$RD\left(\frac{EP^*}{P}\right) = RS$$

(+)

$E \uparrow$ DEPRECIATION OF
HOME CURRENCY

$Y \uparrow$ AND $Y^* \downarrow$

Model 3 equilibrium

Determination of the Long-Run Real Exchange Rate

$$\left. \begin{aligned} q &= q^1 \\ q &= \frac{E P^*}{P} \end{aligned} \right\} \text{DETERMINE } E$$

P, P^* DETERMINED
IN MONEY MKT

$$E = q^1 (P/P^*)$$

